

ADVANCED MATHEMATICAL PHYSICS

PHYS 721

5.1.8 Expand the function $\sin \pi x$ in a series of functions φ_i that are orthogonal (but not normalized) on the range $0 \leq x \leq 1$ when the scalar product has definition

$$\langle f | g \rangle = \int_0^1 f^*(x) g(x) dx.$$

Keep the first four terms of the expansion. The first four φ_i are:

$$\varphi_0 = 1, \quad \varphi_1 = 2x - 1, \quad \varphi_2 = 6x^2 - 6x + 1, \quad \varphi_3 = 20x^3 - 30x^2 + 12x - 1.$$

Goal : Find a_n s.t. $\sin \pi x = \sum_{n=0}^3 a_n \varphi_n$

Solution

for $f(x) = \sum_n a_n \varphi_n$

$$\langle \varphi_m | f \rangle = \sum_n a_n \langle \varphi_m | \varphi_n \rangle = \sum_n a_n \delta_{nm} \langle \varphi_m | \varphi_n \rangle = a_m \langle \varphi_m | \varphi_m \rangle$$

Therefore, the coefficients a_n are just $\frac{\langle \varphi_n | f \rangle}{\langle \varphi_n | \varphi_n \rangle}$

The $\langle \varphi_n | f \rangle$ terms are a linear combination of $\int_0^1 x^n \sin \pi x dx$ terms.

The $\langle \varphi_n | \varphi_n \rangle$ terms are a linear combination of $\int_0^1 x^n dx$ terms

We first find solutions for both integrals. Then, we find the scalar products.

By parts table :

u			dv
x^3	x^2	x	$\sin \pi x$
$3x^2$	$2x$	1	$-\frac{1}{\pi} \cos \pi x$
$6x$	2		$-\frac{1}{\pi^2} \sin \pi x$
6			$\frac{1}{\pi^3} \cos \pi x$
			$\frac{1}{\pi^4} \sin \pi x$

$\int_0^1 x^n \sin \pi x dx :$

$n=0$ $\int_0^1 \sin \pi x dx = -\frac{1}{\pi} \cos \pi x \Big|_0^1 = \frac{2}{\pi}$

$n=1$ $\int_0^1 x \sin \pi x dx = -\frac{1}{\pi} x \cos \pi x + \frac{1}{\pi^2} \sin \pi x \Big|_0^1 = \frac{1}{\pi}$

$n=2$ $\int_0^1 x^2 \sin \pi x dx = -\frac{1}{\pi} x^2 \cos \pi x + \frac{2}{\pi^2} x \sin \pi x + \frac{2}{\pi^3} \cos \pi x \Big|_0^1 = \frac{1}{\pi} - \frac{4}{\pi^3}$

$n=3$ $\int_0^1 x^3 \sin \pi x dx = -\frac{1}{\pi} x^3 \cos \pi x + \frac{3}{\pi^2} x^2 \sin \pi x + \frac{6}{\pi^3} x \cos \pi x - \frac{6}{\pi^4} \sin \pi x \Big|_0^1 = \frac{1}{\pi} - \frac{12}{\pi^3}$

$\int_0^1 x^n dx :$

$n=1$ $\int_0^1 x dx = \frac{x^2}{2} \Big|_0^1 = \frac{1}{2}$

$n=2$ $\int_0^1 x^2 dx = \frac{x^3}{3} \Big|_0^1 = \frac{1}{3}$

$n=N$ $\int_0^1 x^N dx = \frac{x^{N+1}}{N+1} \Big|_0^1 = \frac{1}{N+1}$

Scalar Products	Integrals
$\langle \psi_0 f \rangle$	$\int_0^1 1 \cdot \sin \pi x \, dx = \frac{2}{\pi}$
$\langle \psi_1 f \rangle$	$\int_0^1 (2x-1) \sin \pi x \, dx = 2\left(\frac{1}{\pi}\right) - \frac{2}{\pi} = 0$
$\langle \psi_2 f \rangle$	$\int_0^1 (6x^2 - 6x + 1) \sin \pi x \, dx = 6\left(\frac{1}{\pi} - \frac{4}{\pi^3}\right) - 6\left(\frac{1}{\pi}\right) + \frac{2}{\pi} = \frac{2}{\pi} - \frac{24}{\pi^3}$
$\langle \psi_3 f \rangle$	$\int_0^1 (20x^3 - 30x^2 + 12x - 1) \sin \pi x \, dx = 20\left(\frac{1}{\pi} - \frac{12}{\pi^3}\right) - 30\left(\frac{1}{\pi} - \frac{4}{\pi^3}\right) + 12\left(\frac{1}{\pi}\right) - \frac{2}{\pi} = 0$
$\langle \psi_0 \psi_0 \rangle$	$\int_0^1 1 \cdot 1 \, dx = 1$
$\langle \psi_2 \psi_2 \rangle$	$\int_0^1 (6x^2 - 6x + 1)^2 \, dx = \int_0^1 36x^4 - 72x^3 + 48x^2 - 12x + 1 \, dx = \frac{36}{5} - \frac{72}{4} + \frac{48}{3} - \frac{12}{2} + 1 = \frac{1}{5}$

We finally have that :

$$\begin{aligned}
 \sin \pi x &\approx a_0 \psi_0 + \cancel{a_1 \psi_1} + a_2 \psi_2 + \cancel{a_3 \psi_3} \\
 &= \frac{\langle \psi_0 | \sin \pi x \rangle}{\langle \psi_0 | \psi_0 \rangle} \psi_0 + \frac{\langle \psi_2 | \sin \pi x \rangle}{\langle \psi_2 | \psi_2 \rangle} \psi_2 \\
 &= \frac{2}{\pi} - \frac{\left(\frac{2}{\pi} - \frac{24}{\pi^3}\right)}{1/5} (6x^2 - 6x + 1)
 \end{aligned}$$

5.1.9 Expand the function e^{-x} in Laguerre polynomials $L_n(x)$, which are orthonormal on the range $0 \leq x < \infty$ with scalar product

$$\langle f|g \rangle = \int_0^{\infty} f^*(x)g(x)e^{-x} dx.$$

Keep the first four terms of the expansion. The first four $L_n(x)$ are

$$L_0 = 1, \quad L_1 = 1 - x, \quad L_2 = \frac{2 - 4x + x^2}{2}, \quad L_3 = \frac{6 - 18x + 9x^2 - x^3}{6}.$$

Goal : Find a_n s.t. $e^{-x} = \sum_{n=0}^3 a_n L_n(x)$

Solution

From the previous question, we have that $a_n = \frac{\langle L_n | e^{-x} \rangle}{\langle L_n | L_n \rangle} = \langle L_n | e^{-x} \rangle$ since L_n 's are normal.

The $\langle L_n | e^{-x} \rangle$ terms are just linear combinations of $\int_0^{\infty} x^n e^{-2x} dx$ terms.

$$\int_0^{\infty} x^n e^{-2x} dx :$$

	u		dv
x^3	x^2	x	e^{-2x}
$3x^2$	$2x$	1	$-\frac{1}{2}e^{-2x}$
$6x$	2		$-\frac{1}{4}e^{-2x}$
6			$-\frac{1}{8}e^{-2x}$
			$-\frac{1}{16}e^{-2x}$

$n=0$	$\int_0^{\infty} e^{-2x} dx = -\frac{1}{2}e^{-2x} \Big _0^{\infty} = \frac{1}{2}$
$n=1$	$\int_0^{\infty} x e^{-2x} dx = -\frac{1}{2}x e^{-2x} - \frac{1}{4}e^{-2x} \Big _0^{\infty} = \frac{1}{4}$
$n=2$	$\int_0^{\infty} x^2 e^{-2x} dx = -\frac{1}{2}x^2 e^{-2x} - \frac{1}{2}x e^{-2x} - \frac{1}{4}e^{-2x} \Big _0^{\infty} = \frac{1}{4}$
$n=3$	$\int_0^{\infty} x^3 e^{-2x} dx = -\frac{1}{2}x^3 e^{-2x} - \frac{3}{4}x^2 e^{-2x} - \frac{3}{4}x e^{-2x} - \frac{3}{8}e^{-2x} \Big _0^{\infty} = \frac{3}{8}$

Coefficients	Scalar Products	Integrals
a_0	$\langle L_0 f \rangle$	$\int_0^{\infty} 1 \cdot e^{-2x} dx = \frac{1}{2}$
a_1	$\langle L_1 f \rangle$	$\int_0^{\infty} (1-x) e^{-2x} dx = \frac{1}{2} - \frac{1}{4} = \frac{1}{4}$
a_2	$\langle L_2 f \rangle$	$\int_0^{\infty} \frac{2-4x+x^2}{2} \cdot e^{-2x} dx = \frac{1}{2} - 2\left(\frac{1}{4}\right) + \frac{1}{2}\left(\frac{1}{4}\right) = \frac{1}{8}$
a_3	$\langle L_3 f \rangle$	$\int_0^{\infty} \frac{6-18x+9x^2-x^3}{6} \cdot e^{-2x} dx = \frac{1}{2} - 3\left(\frac{1}{4}\right) + \frac{3}{2}\left(\frac{1}{4}\right) - \frac{1}{6}\left(\frac{3}{8}\right) = \frac{1}{16}$

Therefore, we can write e^{-x} as :

$$e^{-x} \approx \frac{1}{2} + \frac{1}{4}(1-x) + \frac{1}{8} \frac{2-4x+x^2}{2} + \frac{1}{16} \frac{6-18x+9x^2-x^3}{6}$$

5.2.2 Apply the Gram-Schmidt procedure to form the first three Laguerre polynomials:

$$u_n(x) = x^n, \quad n = 0, 1, 2, \dots, \quad 0 \leq x < \infty, \quad w(x) = e^{-x}.$$

The conventional normalization is

$$\int_0^{\infty} L_m(x) L_n(x) e^{-x} dx = \delta_{mn}.$$

$$\text{ANS. } L_0 = 1, \quad L_1 = (1 - x), \quad L_2 = \frac{2 - 4x + x^2}{2}.$$

Let's recall the Gram-Schmidt procedure.

We can generate an orthonormal basis by applying GS on a general basis set.

Let χ_n be a basis set not necessarily orthogonal. Then, ψ_n forms an orthonormal basis by

$$\psi_n = \chi_n - \sum_{\mu=0}^{n-1} \langle \psi_{\mu} | \chi_n \rangle \psi_{\mu} \quad \text{orthogonalizing step}$$

$$\psi_n = \frac{\psi_n}{\langle \psi_n | \psi_n \rangle^{1/2}} \quad \text{normalizing step}$$

Goal is find the first 3 Laguerre polynomials.

n^{th} basis term	General Basis χ_n	Orthogonalize (ψ_n)	Normalize (ψ_n)
$n=0$	$\chi_0 = 1$	$\psi_0 = \chi_0 - \sum_{n=0}^0 \langle \psi_n \chi_0 \rangle \psi_n = 1$	$\psi_0 = \frac{1}{\langle 1 1 \rangle^{1/2}} = \frac{1}{\left[\int_0^{\infty} e^{-x} dx \right]^{1/2}} = 1$
$n=1$	$\chi_1 = x$	$\psi_1 = \chi_1 - \langle \psi_0 \chi_1 \rangle \psi_0$ $= x - \int_0^{\infty} x e^{-x} dx \cdot 1$ $= x - 1$	$\psi_1 = \frac{x-1}{\langle x-1 x-1 \rangle^{1/2}} = \frac{x-1}{\left[\int_0^{\infty} (x-1)^2 e^{-x} dx \right]^{1/2}}$ $= \frac{x-1}{(2-2+1)^{1/2}} = x-1$
$n=2$	$\chi_2 = x^2$	$\psi_2 = \chi_2 - \langle \psi_1 \chi_2 \rangle \psi_1 - \langle \psi_0 \chi_2 \rangle \psi_0$ $= x^2 - \int_0^{\infty} x^2 (1-x) e^{-x} dx \cdot (1-x) - \int_0^{\infty} x^2 e^{-x} dx \cdot 1$ $= x^2 - \int_0^{\infty} (x^2 - x^3) e^{-x} dx \cdot (1-x) - 2$ $= x^2 - (2-6)(1-x) - 2$ $= x^2 - 4x + 2$	$\psi_2 = \frac{x^2 - 4x + 2}{\langle x^2 - 4x + 2 x^2 - 4x + 2 \rangle^{1/2}}$ $= \frac{x^2 - 4x + 2}{\left[\int_0^{\infty} (x^2 - 4x + 2)^2 e^{-x} dx \right]^{1/2}}$ $= \frac{x^2 - 4x + 2}{\left[4! - 8(3!) + 20(2!) - 16(1) + 4 \right]^{1/2}}$ $= \frac{x^2 - 4x + 2}{2}$

Reference Integrals :

$$\int_0^{\infty} x^n e^{-x} dx$$

$$\begin{aligned} n=0 & \int_0^{\infty} e^{-x} dx = -e^{-x} \Big|_0^{\infty} = 1 \\ n=1 & \int_0^{\infty} x e^{-x} dx = -x e^{-x} - e^{-x} \Big|_0^{\infty} = 1 \\ n=2 & \int_0^{\infty} x^2 e^{-x} dx = -x^2 e^{-x} - 2x e^{-x} - 2e^{-x} \Big|_0^{\infty} = 2 \\ n=3 & \int_0^{\infty} x^3 e^{-x} dx = -x^3 e^{-x} - 3x^2 e^{-x} - 6x e^{-x} - 6e^{-x} \Big|_0^{\infty} = 6 \\ n=N & \int_0^{\infty} x^N e^{-x} dx = N! \end{aligned}$$

u			dv
x^4	x^3	x^2	x
$4x^3$	$3x^2$	$2x$	1
$12x^2$	$6x$	2	
$24x$	6		
24			

$$\int_{-\infty}^{\infty} x^n e^{-x^2} dx$$

This is a very interesting integral that requires cool techniques to solve.

Integration by parts yields a clean formula. However, since we are in a physics course, why not do Feynman's Technique?

For Feynman's Technique to work, let's solve the generalized problem $\int_{-\infty}^{\infty} x^n e^{-ax^2} dx$

For $n=0$, $\int_{-\infty}^{\infty} e^{-ax^2} dx$ is just a gaussian with variance $\sigma^2 = \frac{1}{2a}$!

The normalization constant is $\frac{1}{\sqrt{2\pi\sigma^2}} = \frac{\sqrt{a}}{\sqrt{\pi}}$. Therefore, $\int_{-\infty}^{\infty} e^{-ax^2} dx = \sqrt{\frac{\pi}{a}}$

For $n = \text{odd } \mathbb{N}$, $\int_{-\infty}^{\infty} \underbrace{x^n}_{\text{odd}} \underbrace{e^{-ax^2}}_{\text{even}} dx = 0$ since the integrand is an odd function.

For $n = \text{even } \mathbb{N}$, we can generate the solution by Feynman's Technique.

$$\begin{aligned} \int_{-\infty}^{\infty} e^{-ax^2} dx &= \sqrt{\pi} a^{-1/2} \\ \frac{d}{da} \int_{-\infty}^{\infty} x^2 e^{-ax^2} dx &= -\frac{\sqrt{\pi}}{2} a^{-3/2} \\ \frac{d^2}{da^2} \int_{-\infty}^{\infty} x^4 e^{-ax^2} dx &= \frac{\sqrt{\pi}}{2} \cdot \frac{3}{2} a^{-5/2} \\ \vdots \\ \frac{d^k}{da^k} \int_{-\infty}^{\infty} x^{2k} e^{-ax^2} dx &= \sqrt{\pi} \cdot \frac{2k-1}{2} \cdot \frac{2k-3}{2} \cdots \frac{1}{2} a^{-\frac{2k+1}{2}} \\ &= \sqrt{\pi} \frac{(2k-1)!!}{2^k} a^{-\frac{2k+1}{2}} \end{aligned}$$

For $a=1$, we have that $\int_{-\infty}^{\infty} x^{2k} e^{-x^2} dx = \sqrt{\pi} \frac{(2k-1)!!}{2^k}$.

5.2.4 Using the Gram-Schmidt orthogonalization procedure, construct the lowest three Hermite polynomials:

$$u_n(x) = x^n, \quad n = 0, 1, 2, \dots, \quad -\infty < x < \infty, \quad w(x) = e^{-x^2}.$$

For this set of polynomials the usual normalization is

$$\int_{-\infty}^{\infty} H_m(x) H_n(x) w(x) dx = \delta_{mn} 2^m m! \pi^{1/2}.$$

$$\text{ANS. } H_0 = 1, \quad H_1 = 2x, \quad H_2 = 4x^2 - 2.$$

Goal is find the first 3 Hermite polynomials.

n^{th} basis term	General Basis χ_n	Orthogonalize (ψ_n)	Normalize (φ_n)
$n=0$	$\chi_0 = 1$	$\psi_0 = \chi_0 - \sum_{n=0}^0 \langle \psi_n \chi_0 \rangle \psi_n = 1$	$\varphi_0 = \frac{1}{\langle 1 1 \rangle^{1/2}} = \frac{1}{\left[\frac{1}{2^0 0! \pi^{1/2}} \int_{-\infty}^{\infty} e^{-x^2} dx \right]^{1/2}} = 1$
$n=1$	$\chi_1 = x$	$\begin{aligned} \psi_1 &= \chi_1 - \langle \psi_0 \chi_1 \rangle \psi_0 \\ &= x - \frac{1}{2^1 1! \pi^{1/2}} \int_{-\infty}^{\infty} x e^{-x^2} dx \cdot 1 \\ &= x + \frac{1}{4\sqrt{\pi}} \left. e^{-x^2} \right _{-\infty}^{\infty} = x \end{aligned}$	$\begin{aligned} \varphi_1 &= \frac{x}{\langle x x \rangle^{1/2}} = \frac{x}{\left[\frac{1}{2^1 1! \pi^{1/2}} \int_{-\infty}^{\infty} x^2 e^{-x^2} dx \right]^{1/2}} \\ &= \frac{x}{\left(\frac{1}{2\sqrt{\pi}} \cdot \frac{\sqrt{\pi}}{2} \right)^{1/2}} = 2x \end{aligned}$
$n=2$	$\chi_2 = x^2$	$\begin{aligned} \psi_2 &= \chi_2 - \langle \psi_1 \chi_2 \rangle \psi_1 - \langle \psi_0 \chi_2 \rangle \psi_0 \\ &= x^2 - \frac{\int_{-\infty}^{\infty} 2x^3 e^{-x^2} dx \cdot 2x}{2^1 1! \pi^{1/2}} - \frac{\int_{-\infty}^{\infty} x^2 e^{-x^2} dx \cdot 1}{2^0 0! \pi^{1/2}} \\ &= x^2 - \frac{\pi^{1/2}}{2 \pi^{1/2}} \\ &= x^2 - \frac{1}{2} \end{aligned}$	$\begin{aligned} \varphi_2 &= \frac{x^2 - 1/2}{\langle x^2 - 1/2 x^2 - 1/2 \rangle^{1/2}} = \frac{x^2 - 1/2}{\left[\frac{1}{2^2 2! \pi^{1/2}} \int_{-\infty}^{\infty} (x^2 - 1/2)^2 e^{-x^2} dx \right]^{1/2}} \\ &= \frac{x^2 - 1/2}{\left[\frac{1}{8\sqrt{\pi}} \left(\frac{3\sqrt{\pi}}{4} - \frac{\sqrt{\pi}}{2} + \frac{\sqrt{\pi}}{4} \right) \right]^{1/2}} \\ &= 4(x^2 - 1/2) = 4x^2 - 2 \end{aligned}$

5.2.8 Form a set of three orthonormal vectors by the Gram-Schmidt process using these input vectors in the order given:

$$\mathbf{c}_1 = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}, \quad \mathbf{c}_2 = \begin{pmatrix} 1 \\ 1 \\ 2 \end{pmatrix}, \quad \mathbf{c}_3 = \begin{pmatrix} 1 \\ 0 \\ 2 \end{pmatrix}.$$

Finally, vectors! Something I can get behind.

The GS process is identical but now with vectors instead of functions.

h basis term	General Basis χ_n	Orthogonalize (Ψ_n)	Normalize (Ψ_n)
$n=0$	c_0	$a_0 = c_0 - \sum_{n=0}^0 \langle a_n c_0 \rangle a_n = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$	$b_0 = \frac{a_0}{ a_0 } = \frac{1}{\sqrt{3}} \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$
$n=1$	c_1	$a_1 = c_1 - \langle b_0 c_1 \rangle b_0$ $= \begin{pmatrix} 1 \\ 1 \\ 2 \end{pmatrix} - \frac{1}{\sqrt{3}} \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} \cdot \begin{pmatrix} 1 \\ 1 \\ 2 \end{pmatrix} \frac{1}{\sqrt{3}} \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$ $= \begin{pmatrix} 1 \\ 1 \\ 2 \end{pmatrix} - \frac{4}{3} \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$ $= \frac{1}{3} \begin{pmatrix} -1 \\ -1 \\ 2 \end{pmatrix}$	$b_1 = \frac{a_1}{ a_1 } = \frac{1}{\sqrt{6}} \begin{pmatrix} -1 \\ -1 \\ 2 \end{pmatrix}$
$n=2$	c_2	$a_2 = c_2 - \langle b_0 c_2 \rangle b_0 - \langle b_1 c_2 \rangle b_1$ $= \begin{pmatrix} 1 \\ 0 \\ 2 \end{pmatrix} - \frac{1}{\sqrt{3}} \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} \cdot \begin{pmatrix} 1 \\ 0 \\ 2 \end{pmatrix} \frac{1}{\sqrt{3}} \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} - \frac{1}{\sqrt{6}} \begin{pmatrix} -1 \\ -1 \\ 2 \end{pmatrix} \cdot \begin{pmatrix} 1 \\ 0 \\ 2 \end{pmatrix} \frac{1}{\sqrt{6}} \begin{pmatrix} -1 \\ -1 \\ 2 \end{pmatrix}$ $= \begin{pmatrix} 1 \\ 0 \\ 2 \end{pmatrix} - \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} - \frac{1}{2} \begin{pmatrix} -1 \\ -1 \\ 2 \end{pmatrix}$ $= \frac{1}{2} \begin{pmatrix} -1 \\ -1 \\ 0 \end{pmatrix}$	$b_2 = \frac{a_2}{ a_2 } = \frac{1}{\sqrt{2}} \begin{pmatrix} -1 \\ -1 \\ 0 \end{pmatrix}$

5.3.2 U and V are two arbitrary operators. Without introducing matrix representations of these operators, show that

$$(UV)^\dagger = V^\dagger U^\dagger.$$

Note the resemblance to adjoint matrices.

Let U, V be two arbitrary operators on the Hilbert space $\mathcal{H}$ such that $\forall$
 $\langle f | Ag \rangle = \langle A^\dagger f | g \rangle$ for some A on $\mathcal{H}$

Then,

$$\underbrace{\langle (UV)^\dagger f | g \rangle}_{\text{def of adjoint of } UV} = \underbrace{\langle f | UVg \rangle}_{\text{def of adjoint of } U} = \underbrace{\langle U^\dagger f | Vg \rangle}_{\text{def of adjoint of } V} = \langle V^\dagger U^\dagger f | g \rangle$$

$$\therefore (UV)^\dagger = V^\dagger U^\dagger$$

5.3.3 Consider a Hilbert space spanned by the three functions $\varphi_1 = x_1$, $\varphi_2 = x_2$, $\varphi_3 = x_3$, and a scalar product defined by $\langle x_\nu | x_\mu \rangle = \delta_{\nu\mu}$.

(a) Form the 3×3 matrix of each of the following operators:

$$A_1 = \sum_{i=1}^3 x_i \left(\frac{\partial}{\partial x_i} \right), \quad A_2 = x_1 \left(\frac{\partial}{\partial x_2} \right) - x_2 \left(\frac{\partial}{\partial x_1} \right).$$

(b) Form the column vector representing $\psi = x_1 - 2x_2 + 3x_3$.

(c) Form the matrix equation corresponding to $\chi = (A_1 - A_2)\psi$ and verify that the matrix equation reproduces the result obtained by direct application of $A_1 - A_2$ to ψ .

Let's recall the basis expansion of operators :

Let A be an operator on Hilbert space spanned by φ_i for $i=1,2,3$.

Then, $A\varphi_j = \sum_{i=1}^3 A_{ij} \varphi_i$

And by taking $\langle \varphi_k |$ on both sides :

$$\langle \varphi_k | A \varphi_j \rangle = \sum_{i=1}^3 A_{ij} \langle \varphi_k | \varphi_i \rangle$$

$$= A_{kj} \quad \therefore \text{Elements of } A \text{ is } A_{ij} = \langle \varphi_i | A \varphi_j \rangle$$

a) Find A_1 such that $A_1 = \sum_{i=1}^3 x_i \frac{\partial}{\partial x_i}$

Solution :

$$A_{ij} = \langle \varphi_i | A \varphi_j \rangle$$

$$= \langle x_i | \sum_{k=1}^3 x_k \frac{\partial}{\partial x_k} x_j \rangle$$

$$= \sum_{k=1}^3 \langle x_i | x_k \frac{\partial x_j}{\partial x_k} \rangle = \sum_{k=1}^3 \langle x_i | x_k \delta_{jk} \rangle = \langle x_i | x_j \rangle = \delta_{ij}$$

$$\Rightarrow A_1 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Find A_2 such that $A_2 = x_1 \frac{\partial}{\partial x_2} - x_2 \frac{\partial}{\partial x_1}$

Solution :

$$A_{ij} = \langle \varphi_i | A \varphi_j \rangle$$

$$= \langle x_i | [x_1 \frac{\partial}{\partial x_2} - x_2 \frac{\partial}{\partial x_1}] x_j \rangle$$

$$= \langle x_i | x_1 \delta_{2j} - x_2 \delta_{1j} \rangle$$

$$= \delta_{2j} \langle x_i | x_1 \rangle - \delta_{1j} \langle x_i | x_2 \rangle = \delta_{2j} \delta_{i1} - \delta_{1j} \delta_{i2} \Rightarrow \begin{cases} 1 & \text{for } (i,j) = (1,2) \\ -1 & \text{for } (i,j) = (2,1) \\ 0 & \text{otherwise} \end{cases}$$

$$\Rightarrow A_2 = \begin{bmatrix} 0 & 1 & 0 \\ -1 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

b) Simply :

$$\psi = \begin{pmatrix} 1 \\ -2 \\ 3 \end{pmatrix} \cdot \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} \quad \text{i.e.} \quad \psi = \begin{pmatrix} 1 \\ -2 \\ 3 \end{pmatrix}$$

c) Let's first apply the operation $A_1 - A_2$ on ψ then verify using matrix operations.

Direct Application :

$$\begin{aligned} (A_1 - A_2) \psi &= \left[\sum_{i=1}^3 x_i \partial_{x_i} - x_1 \partial_{x_2} + x_2 \partial_{x_1} \right] (x_1 - 2x_2 + 3x_3) \\ &= \underbrace{x_1 + x_2}_{(A_1 - A_2)x_1} - \underbrace{2x_2 + 2x_1}_{(A_1 - A_2) - 2x_2} + \underbrace{3x_3}_{(A_1 - A_2) 3x_3} = 3x_1 - x_2 + 3x_3 \quad \text{or} \quad \begin{pmatrix} 3 \\ -1 \\ 3 \end{pmatrix} \end{aligned}$$

Matrix Operations :

$$\begin{aligned} (A_1 - A_2) \psi &= \left(\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} - \begin{bmatrix} 0 & 1 & 0 \\ -1 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \right) \begin{pmatrix} 1 \\ -2 \\ 3 \end{pmatrix} \\ &= \begin{bmatrix} 1 & -1 & 0 \\ 1 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{pmatrix} 1 \\ -2 \\ 3 \end{pmatrix} \\ &= \begin{pmatrix} 3 \\ -1 \\ 3 \end{pmatrix} \quad \text{as required} \quad \square \end{aligned}$$

* Another simple way to get matrices A_1 and A_2 is by applying them directly on the basis functions.

$$\begin{aligned} A_1 x_1 &= x_1 = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} & \therefore A_1 &= \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \\ A_1 x_2 &= x_2 = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} \\ A_1 x_3 &= x_3 = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} \end{aligned}$$

$$\begin{aligned} A_2 x_1 &= -x_2 = \begin{pmatrix} 0 \\ -1 \\ 0 \end{pmatrix} & \therefore A_2 &= \begin{pmatrix} 0 & 1 & 0 \\ -1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} \\ A_2 x_2 &= x_1 = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} \\ A_2 x_3 &= 0 = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix} \end{aligned}$$

- 5.3.4** (a) Obtain the matrix representation of $A = x(d/dx)$ in a basis of Legendre polynomials, keeping terms through P_3 . Use the orthonormal forms of these polynomials as given in 5.2.1 and the scalar product defined there.
- (b) Expand x^3 in the orthonormal Legendre polynomial basis.
- (c) Verify that Ax^3 is given correctly by its matrix representation.

Let's recall the first 4 orthonormal Legendre Polynomials :

$$\psi_0 = \frac{1}{\sqrt{2}}, \quad \psi_1 = \sqrt{\frac{3}{2}} x, \quad \psi_2 = \sqrt{\frac{5}{2}} \left(\frac{3}{2} x^2 - \frac{1}{2} \right), \quad \psi_3 = \sqrt{\frac{7}{2}} \left(\frac{5}{2} x^3 - \frac{3}{2} x \right)$$

Awesome! Goal is Find matrix A s.t. $A = x \frac{d}{dx}$

Solution :

We apply A on each of our ψ_i . Then, write the result as a component of the ψ_i basis.

$$A\psi_0 = x \frac{d}{dx} \left(\frac{1}{\sqrt{2}} \right) = 0 = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 0 \end{pmatrix}$$

$$A\psi_1 = x \frac{d}{dx} \left(\sqrt{\frac{3}{2}} x \right) = \sqrt{\frac{3}{2}} x = \begin{pmatrix} 0 \\ 1 \\ 0 \\ 0 \end{pmatrix}$$

$$A\psi_2 = x \frac{d}{dx} \left(\sqrt{\frac{5}{2}} \left(\frac{3}{2} x^2 - \frac{1}{2} \right) \right) = \sqrt{\frac{5}{2}} 3x^2 = 2\psi_2 + \sqrt{\frac{5}{2}} = 2\psi_2 + \sqrt{5}\psi_0 = \begin{pmatrix} \sqrt{5} \\ 0 \\ 2 \\ 0 \end{pmatrix}$$

$$A\psi_3 = x \frac{d}{dx} \left(\sqrt{\frac{7}{2}} \left(\frac{5}{2} x^3 - \frac{3}{2} x \right) \right) = \sqrt{\frac{7}{2}} \left(3 \cdot \frac{5}{2} x^2 - \frac{3}{2} \right) = 3\psi_3 + 2\sqrt{\frac{7}{2}} \left(\frac{3}{2} x \right) = 3\psi_3 + \sqrt{21} \psi_1 = \begin{pmatrix} 0 \\ \sqrt{21} \\ 0 \\ 3 \end{pmatrix}$$

$$\therefore A = \begin{bmatrix} 0 & 0 & \sqrt{5} & 0 \\ 0 & 1 & 0 & \sqrt{21} \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 3 \end{bmatrix}$$

$A\psi_0 \quad A\psi_1 \quad A\psi_2 \quad A\psi_3$

b) Recall from Section 5.1 :

A function f can be expanded in an arbitrary basis ψ_i by -

$$f = \sum_{i=0}^N \langle \psi_i | f \rangle \psi_i$$

Therefore, we are required to expand $f_{\text{ex}} = x^3$ on ψ_i 's which we will do by calculating scalar products.

Coefficient	Scalar Product	Integral
a_0	$\langle \psi_0 x^3 \rangle$	$\int_{-1}^1 \frac{1}{\sqrt{2}} x^3 dx = 0$ since x^3 is odd.
a_1	$\langle \psi_1 x^3 \rangle$	$\int_{-1}^1 \sqrt{\frac{3}{2}} x^4 dx = \sqrt{\frac{3}{2}} \frac{x^5}{5} \Big _{-1}^1 = \frac{\sqrt{6}}{5}$
a_2	$\langle \psi_2 x^3 \rangle$	$\int_{-1}^1 \sqrt{\frac{5}{2}} \left(\frac{3}{2} x^5 - \frac{1}{2} x^3 \right) dx = 0$ since integrand is odd
a_3	$\langle \psi_3 x^3 \rangle$	$\int_{-1}^1 \sqrt{\frac{7}{2}} \left(\frac{5}{2} x^6 - \frac{3}{2} x^4 \right) dx = \sqrt{\frac{7}{2}} \left(\frac{5}{14} x^7 - \frac{3}{10} x^5 \right) \Big _{-1}^1 = \sqrt{\frac{7}{2}} \left(\frac{5}{7} - \frac{3}{5} \right) = \frac{4}{35} \sqrt{\frac{7}{2}}$

$$\therefore \text{ We have that } x^3 = a_1 \psi_1 + a_3 \psi_3 \rightarrow \begin{pmatrix} 0 \\ \frac{\sqrt{6}}{5} \\ 0 \\ \frac{4}{35} \sqrt{\frac{7}{2}} \end{pmatrix}$$

c) Let's verify Ax^3 by direct computation & matrix operations.

Direct Computation:

$$Ax^3 = x \partial_x (x^3) = 3x^3$$

Matrix Operations:

$$Ax^3 = \begin{bmatrix} 0 & 0 & \sqrt{5} & 0 \\ 0 & 1 & 0 & \sqrt{21} \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 3 \end{bmatrix} \begin{pmatrix} 0 \\ \frac{\sqrt{6}}{5} \\ 0 \\ \frac{4}{35} \sqrt{\frac{7}{2}} \end{pmatrix} = \begin{pmatrix} 0 \\ \frac{\sqrt{6}}{5} + \frac{4}{5} \sqrt{\frac{3}{2}} \\ 0 \\ \frac{12}{35} \sqrt{\frac{7}{2}} \end{pmatrix} = \frac{\sqrt{6}}{5} + \frac{4}{5} \sqrt{\frac{3}{2}} \psi_1 + \frac{12}{35} \sqrt{\frac{7}{2}} \psi_3$$

$$= \left(\frac{\sqrt{6}}{5} + \frac{4}{5} \sqrt{\frac{3}{2}} \right) \left(\sqrt{\frac{3}{2}} x \right) + \frac{12}{35} \sqrt{\frac{7}{2}} \left(\sqrt{\frac{7}{2}} \left(\frac{5}{2} x^3 - \frac{3}{2} x \right) \right)$$

$$= \cancel{\frac{3}{5} x} + \cancel{\frac{6}{5} x} + 3x^3 - \cancel{\frac{9}{5} x}$$

$$= 3x^3 \text{ as required } \blacksquare$$